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Week 10

KI1101 General Chemistry 1

Basics of Chemical Bonding

Chemical Bonds

- Attractive forces that hold atoms together in complex substances
- Molecules and ionic compounds

Why study?

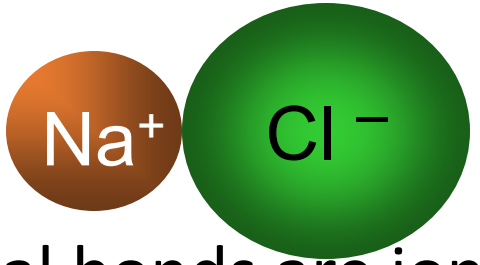
- Changes in these bonding forces are the underlying basis of chemical reactivity
- During reaction:
  - Break old bonds
  - Form new bonds

Classes of Bonds

- **Covalent bonding**
  - Occurs in molecules
  - Sharing of e^-'s
- **Ionic Bonding**
  - Occurs in ionic solid
  - e^-'s transferred from 1 atom to another
  - Simpler
  - We will look at this first

Ionic Bonds

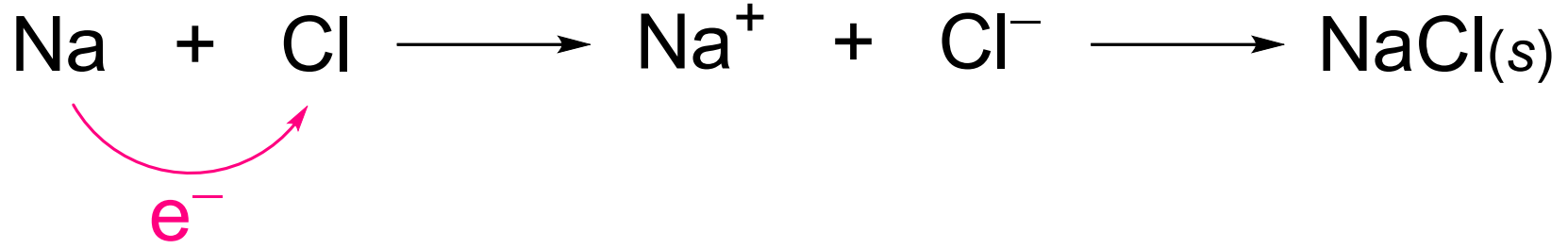
- Result from attractive forces between oppositely charged particles



- Metal - nonmetal bonds are ionic because:
  - Metals have
    - Low ionization energies
    - Easily lose e^- to be stable
  - Non-metals have
    - Very exothermic electron affinities
  - Formation of lattice stabilizes ions

Ionic Compounds

- Formed from metal and nonmetal



http://www.visionlearning.com/library/module_viewer.php?c3=&mid=55&l=

- **Ionic Bond**
  - Attraction between + and – ions in ionic compound.
- **Why does this occur? Why is e^- transferred?**
  - Why Na^+ and not Na^{2+} or Na^-?
  - Why Cl^- and not Cl^{2-} or Cl^+?

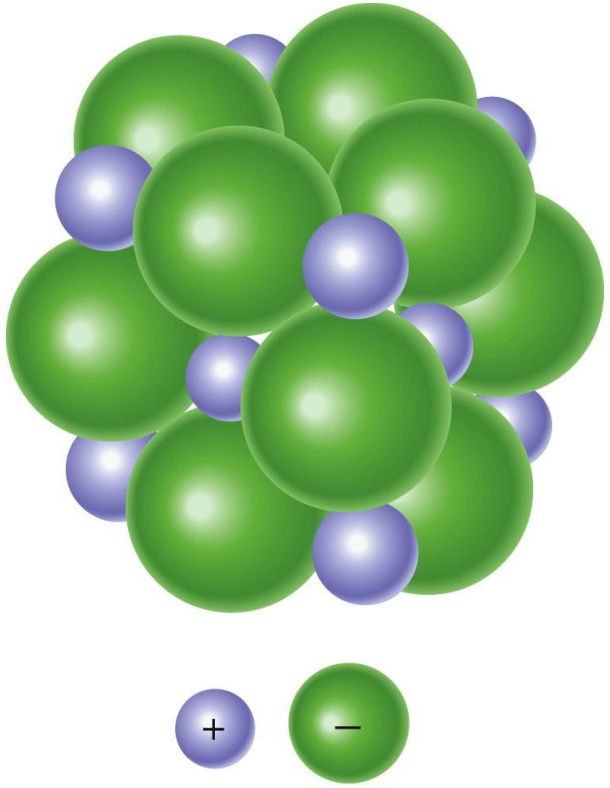
Ionic Compounds

Ionic crystals:

- Exist in 3-dimensional array of cations and anions = **lattice structure**

Ionic chemical formulas:

- Always written as empirical formula
- Smallest whole number ratio of cation to anion



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Energetics

- Must look at energy of system to answer these questions
- For any stable compound to form from its elements
  - Potential Energy of system must be lowered.
  - Net ↓ in energy $\Delta H_f^\circ < 0$ (negative)
- What are factors contributing to energy lowering for ionic compound?
  - Use Hess's Law to determine
  - Conservation of energy
  - Envision two paths

Paths to Evaluate Energy

1. $Na(s) + \frac{1}{2}Cl_2(g) \rightarrow NaCl(s)$ $\quad \Delta H_f^\circ = -411.1$ kJ/mol

2. Stepwise path

$Na(s) \rightarrow Na(g)$ $\quad \Delta H_f^\circ(Na, g) = 107.8$ kJ/mol

$\frac{1}{2}Cl_2(g) \rightarrow Cl(g)$ $\quad \Delta H_f^\circ(Cl, g) = 121.3$ kJ/mol

$Na(g) \rightarrow Na^+(g) + e^-$ $\quad IE(Na) = 495.4$ kJ.mol

$Cl(g) + e^- \rightarrow Cl^-(g)$ $\quad EA(Cl) = -348.8$ kJ/mol

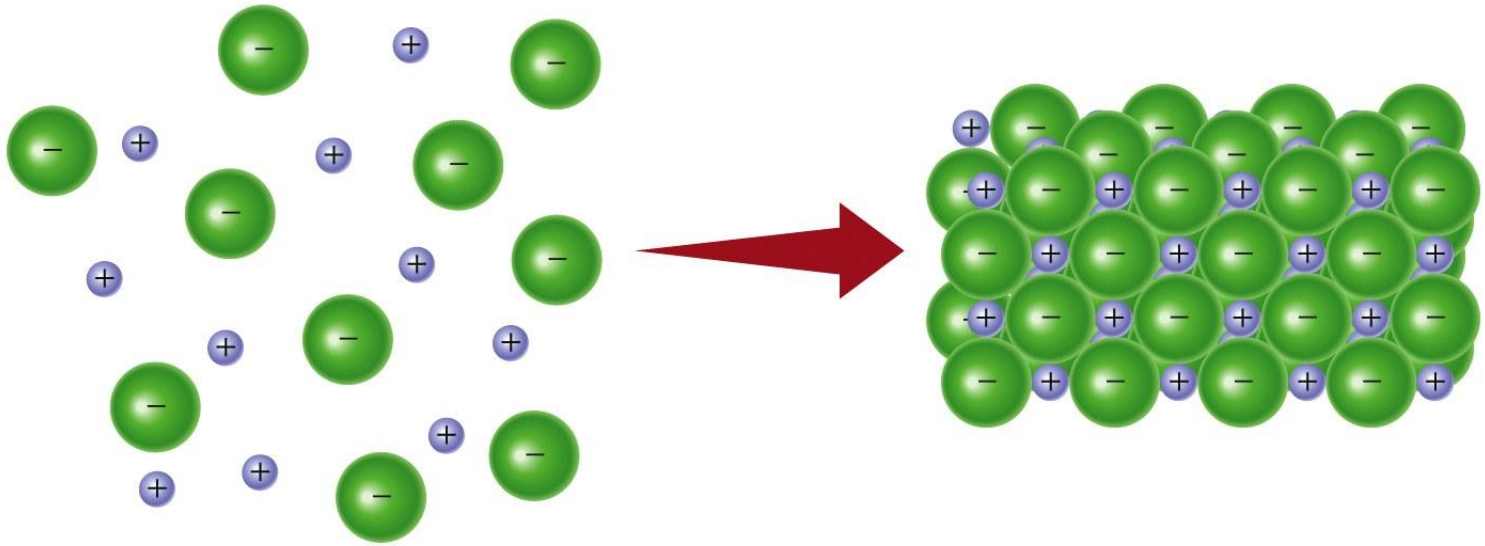
$Na^+(g) + Cl^-(g) \rightarrow NaCl(s)$ $\quad \Delta H_{lattice} = -787$ kJ/mol

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$Na(s) + \frac{1}{2}Cl_2(g) \longrightarrow NaCl(s)$ $\quad \Delta H_f^\circ = -411$ kJ/mol

Lattice Energy

- Amount that PE of system decreases when 1 mole of solid salt is formed from its gas phase ions.
- Energy released when ionic lattice forms.



1 mol Na^+ and 1 mol Cl^-
(gaseous ions from NaCl)

1 mol NaCl
(solid, crystalline NaCl)

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Lattice Energy

- Always $\Delta H_{Lattice}$ = – = exothermic
- $\Delta H_{Lattice}$ gets more exothermic (larger negative value) as ions of opposite charge in crystal lattice are brought closer together as they wish to be.
- Ions tightly packed with opposite charged ions next to each other.

$$\Delta H_{\text{Lattice}} \propto \frac{q_+ q_-}{d}$$

- Any ↑ in PE due to ionizing atoms is more than met by ↓ in PE from formation of crystal lattice. Even for +2 and –2 ions
- ∴ overall exothermic to form ionic solids and they are stable compounds.

Your Turn!

Assuming that the separation between cations and anions in the lattice is nearly identical, which species would have the greatest lattice energy?

A. Sodium chloride

B. Calcium chloride

C. Calcium nitride

D. Sodium oxide

E. Calcium oxide

Why Do Metals form Cations and Nonmetals form Anions

Metal

- Left hand side of Periodic Table
- **IE small** and positive
  - Little energy required to remove **e**$^-$ to give cations
- **EA small** and negative or positive
  - Not favorable to attract an **e**$^-$ to it.
- Least expensive, energy-wise, to form cation

Nonmetal

- Right hand side of Periodic Table
- **IE large** and positive
  - Difficult to remove **e**$^-$
- **EA large** and negative
  - But easy to add **e**$^-$
  - Exothermic—large amount of E given off
  - PE of system ↓
- Least expensive, energy-wise, to form anion

Electron Configurations of Ions

- How **e**$^{-}$'ic structure affects types of ions formed

Ex.

Na $\quad$ **$1s^2\ 2s^2\ 2p^6\ 3s^1$ = [Ne] $3s^1$**

Na^+ $\quad$ **$1s^2\ 2s^2\ 2p^6$ = [Ne]**

IE_1 = 496 kJ/mol $\quad$ small $\quad$ not too difficult

IE_2 = 4563 kJ/mol $\quad$ large $\quad$ ~10 x larger very difficult

- Can remove 1st **e**$^{-}$, as doesn't cost too much
- Can't remove 2nd **e**$^{-}$, as can't regain lost energy from lattice
- $\therefore$ doesn't form.

Electron Configurations of Ions

Ex. Ca [Ar] $4s^2$

Ca^{2+} [Ar]

IE_1	small = 590 kJ/mol	not too difficult
IE_2	small = 1140 kJ/mol	not too difficult
IE_3	large = 4940 kJ/mol	too difficult

- Can regain by lattice energy ~2000 kJ/mole if +2, –2 charges.
- But 3rd **e**$^-$ too hard to remove
- Can't recoup required energy through lattice formation.
- ∴ doesn't happen

Electron Configurations of Ions

- **Stability of Noble Gas Core below valence e^-'s effectively limits # e^-'s that metals lose.**
- **Ions formed have Noble gas e^- configuration**
  - True for anions and cations

Ex. Cl **$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$ = [Ne]$3s^2\ 3p^5$**

Cl^- **$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$ = [Ar]**

- Adding another **e^-**
  - requires putting it into next higher ***n*** shell
- Energy cost too high

Electron Configurations of Ions

Ex.

O $\quad$ **$1s^2\ 2s^2\ 2p^4$**

O^- $\quad$ **$1s^2\ 2s^2\ 2p^5$** $\quad$ **EA_1** = –141 kJ/mol

O^{2-} $\quad$ **$1s^2\ 2s^2\ 2p^6$ = [Ne]** $\quad$ **EA_2** = +844 kJ/mol

EA_{net} = +703 kJ/mol $\quad$ endothermic

- Energy required more than made up for by ↑ in $\Delta H_{Lattice}$ caused by higher –2 charge

Electron Configurations of Ions

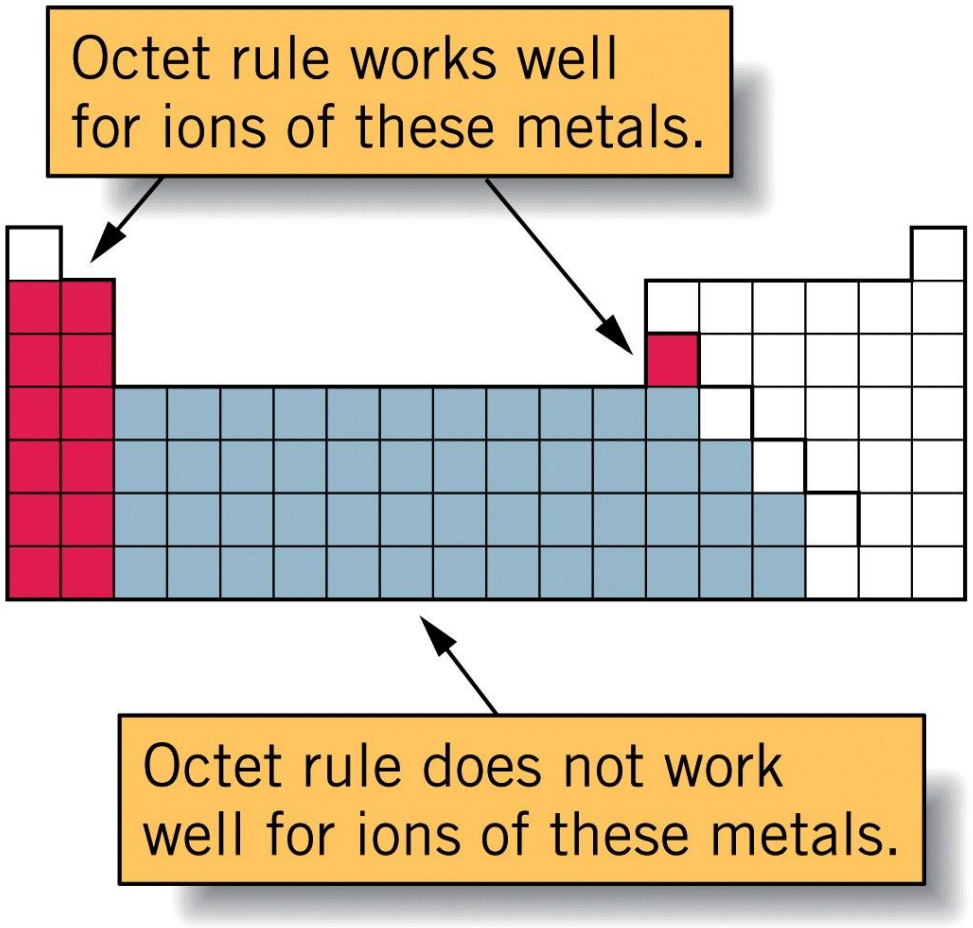
Generalization:

- When ions form
  - Atoms of most representative elements (***s*** and ***p*** block)
  - Tend to gain or lose **e**$^-$'s to obtain nearest Noble gas **e**$^-$ configuration
  - Except He (2 **e**$^-$'s), all noble gases have 8 **e**$^-$'s in highest ***n*** shell

Octet Rule

- Atoms tend to gain or lose **e**$^-$'s until they have achieved outer (valence) shell containing octet (8 **e**$^-$'s)

Octet Rule



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- Works well with
  - Group IA and IIA metals
  - Al
  - Non-metals
- H and He can't obey
  - Limited to 2 $\mathbf{e^-}$'s in $\boldsymbol{n} = \mathbf{1}$ shell
- Doesn't work with
  - Transition metals
  - Post transition metals

Your Turn!

What is the correct electron configuration for Cs and Cs^+ ?

A. **[Xe] $6s^1$, [Xe]**

B. **[Xe] $6s^2$, [Xe] $6s^1$**

C. **[Xe] $5s^1$, [Xe]**

D. **[Xe] $6s^1$, [Xe] $6s^2$**

E. **[Xe] $6p^2$, [Xe] $6p^1$**

Post Transition Metals

Ex.

Sn **[Kr] $4d^{10}$ $5s^2$ $5p^2$**

Sn^{2+} **[Kr] $4d^{10}$ $5s^2$**

- Neither has Noble Gas **e^-** configuration
- Have emptied ***5p*** subshell

Sn^{4+} **[Kr] $4d^{10}$**

- Does have empty ***n* = 5**

Transition Metals

- 1st **e**$^{-}$'s lost from ***ns*** orbital of outer shell
- Lose **e**$^{-}$'s from highest ***n*** first, then highest ***ℓ***

Ex. Fe **[Ar] 3*d*^{6} 4*s*^{2}**

Fe^{2+} **[Ar] 3*d*^{6}** loses 4***s*** **e**$^{-}$'s 1st

Fe^{3+} **[Ar] 3*d*^{5}** then loses 3***d*** **e**$^{-}$

- Extra stability due to exactly half-filled ***d*** subshell.
- ***Consequences***
  - TM^{2+} common oxidation state as remove 2 **e**$^{-}$'s from outer ***ns*** shell
  - Ions of larger charge result from loss of ***d*** **e**$^{-}$'s.

Transition Metals

- Not easy to predict which ions form and which are stable
- But ions with exactly filled or half-filled ***d*** subshells are extra stable and ∴ tend to form.
- Mn^{2+} **[Ar]3*d*^{5}**
- Fe^{3+} **[Ar]3*d*^{5}**
- Zn^{2+} **[Ar]3*d*^{10}**

Your Turn!

What is the correct electronic configuration for Cu and Cu^{2+} ?

A. **[Ar] $3d^{9}4s^{2}$, [Ar] $3d^{9}$**

B. **[Ar] $3d^{10}4s^{1}$, [Ar] $3d^{8}4s^{1}$**

C. **[Ar] $3d^{10}4s^{1}$, [Ar] $3d^{9}$**

D. **[Ar] $3d^{9}4s^{2}$, [Ar] $3d^{10}4s^{1}$**

E. **[K] $3d^{9}4s^{2}$, [Ar] $3d^{9}$**

- Filled and half-filled orbitals are particularly stable.

Predicting Cation Configurations

Consider Bi, whose configuration is $[Xe]6s^2\ 4f^{14}\ 5d^{10}\ 6p^3$. What ions are expected?

Rewrite config'n: $[Xe]4f^{14}\ 5d^{10}\ 6s^2\ 6p^3$

Bi^{3+} and Bi^{5+}

Consider Fe, whose configuration is: $[Ar]4s^2\ 3d^6$ What ions are expected?

Rewrite config'n: $[Ar]3d^6\ 4s^2$

Fe^{2+} and Fe^{3+}

Predicting Anion Configurations

Non-metals gain electrons to become isoelectronic with next larger noble gas

O: $[He]2s^2 2p^4$ + ? e^- → ?

2

O^{2-} : $[He]2s^2\ 2p^6$

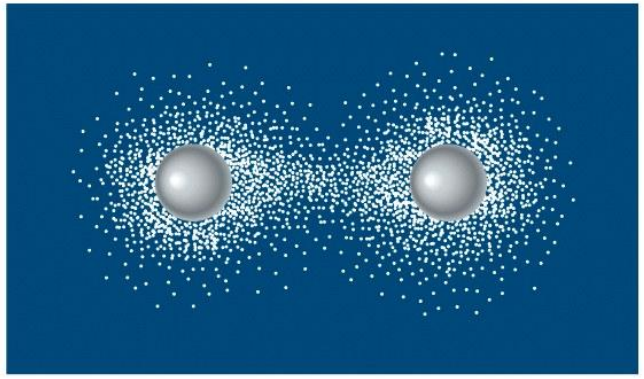
N: $[He]2s^2 2p^3$ + ? e^- → ?

3

N^{3-} : $[He]2s^2\ 2p^6$

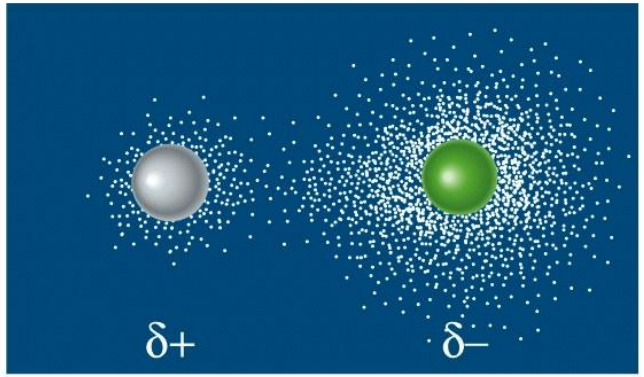
Electronegativity and Bond Polarity

- 2 atoms of same element form bond
  - **Equal** sharing of e⁻'s
- 2 atoms of different elements form bond
  - **Unequal** sharing of e⁻'s



H——H

(*a*)



H——Cl

(*b*)

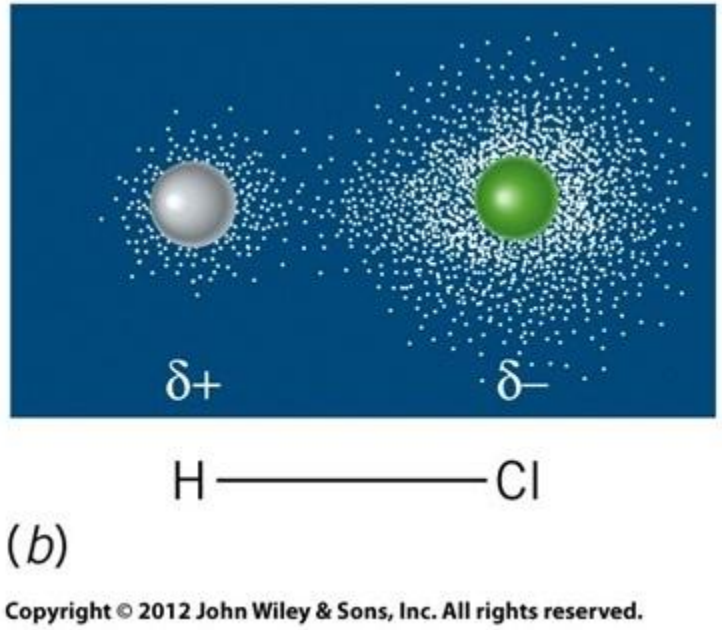
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Why ?

- 1 atom usually attracts e^-s more strongly than other

Result:

- Unbalanced distribution of e^- density within bond
  - e^- cloud **tighter** around Cl in HCl
  - Slight + charge around H
  - Slight – charge around Cl
  - Not complete transfer of e^- from 1 atom to another.



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Electronegativity and Bond Polarity

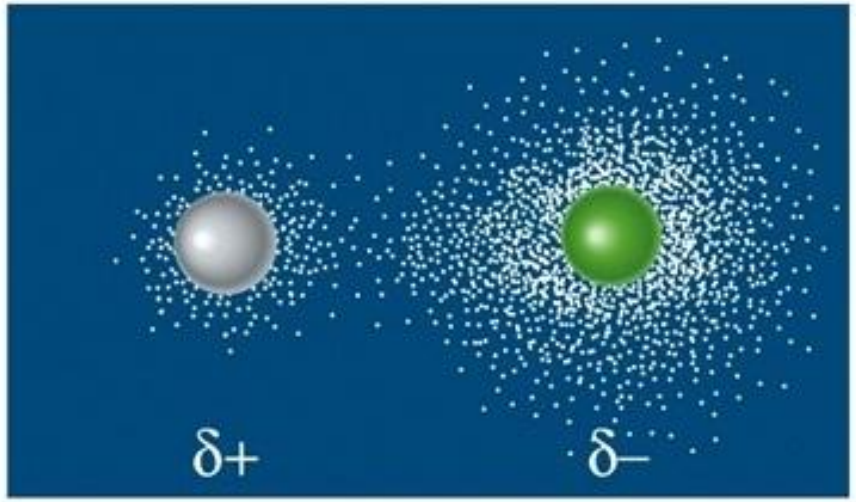
- Leads to concept of **Partial charges**

δ^+ δ^-

H——Cl

δ^+ on H = +0.17

δ^- on Cl = –0.17



(*b*)

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Polar Covalent Bond

- **Aka Polar bond**
- Bond that carries partial + and – charges at opposite ends
- Bond is **dipole**
  - 2 poles or 2 charges involved

Polar Molecule

- Molecule has partial + and – charges at opposite ends due to a polar bond

http://web.visionlearning.com/custom/chemistry/animations/CHE1.7-an-H2Obond.shtml

Dipole Moment

- Quantitative measure of extent to which bond is polarized.
- Dipole moment = Charge on either end × distance between them
- $\boldsymbol{\mu = q \times r}$
- Units = debye (D)
- 1D = 3.34×10^{-30} C·m (Coulombs·meter)
- The **size** of the **dipole moment** or the **degree of polarity** in the bond depends on the **differences** in abilities of bonded atoms to attract e^-'s to themselves

Dipole Moments and Bond Lengths for Some Diatomic Molecules

Table 9.3 Dipole Moments and Bond Lengths for Some Diatomic Molecules[a]

Compound	Dipole Moment (D)	Bond Length (pm)
HF	1.83	91.7
HCl	1.09	127
HBr	0.82	141
HI	0.45	161
CO	0.11	113
NO	0.16	115

[a]Source: National Institute of Standards and Technology.

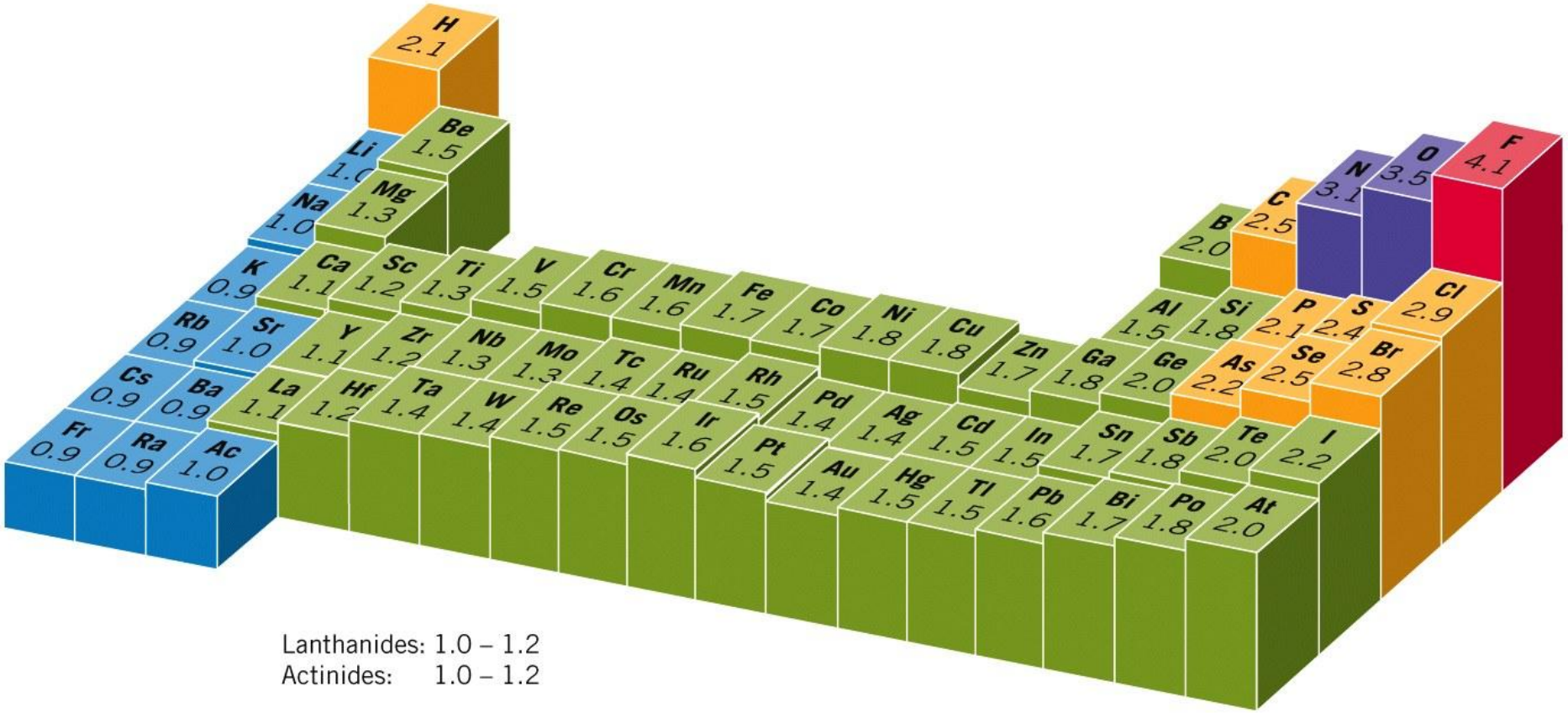
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Electronegativity

- Relative attraction of atom for e^-s in bond
- Ability of bonded atom to attract e^-s to itself
- Quantitative basis
  - Table of electronegativities - fig 8.5
- Difference in electronegativity
  - = estimate of bond polarity
  - $\Delta\chi = |\chi_1 - \chi_2|$
- **Ex.** N—H Si—F

  δ^- δ^+ δ^+ δ^-

Electronegativity Table



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Electronegativity

- Must know for 2nd row and H.
- χ(H) ~ 2.0
- χ(F) ~ 4.0
- $\chi\downarrow$ by 0.5 for each element as go to left

	H						
χ	2						
	Li	**Be**	**B**	**C**	**N**	**O**	**F**
χ	1	1.5	2	2.5	3	3.5	4
					P	**S**	**Cl**
					2	2.5	3.0

Learning Check

- Calculate the bond polarity of the following
- HCl χ(H)~2 χ(Cl)~3 $\Delta\chi = 3 - 2 = 1$
- CO χ(C)~2.5 χ(O)~3.5 $\Delta\chi = 3.5 - 2.5 = 1$
- CH χ(C)~2.5 χ(H)~2 $\Delta\chi = 2.5 - 2 = 0.5$
- LiF χ(Li)~1 χ(F)~4 $\Delta\chi = 4 - 1 = 3$
- BeO χ(Be)~1.5 χ(O)~3.5 $\Delta\chi = 1.5 - 3.5 = 2$

Your Turn!

Which of the following species has the least polar bond?

A. HCl

B. HF

C. HI

D. HBr

Trends in Electronegativity

- $\chi\uparrow$ from left to right
  - across period as $\mathbf{Z_{eff}}\uparrow$
- $\chi\downarrow$ from top to bottom
  - down group as $\boldsymbol{n}\uparrow$

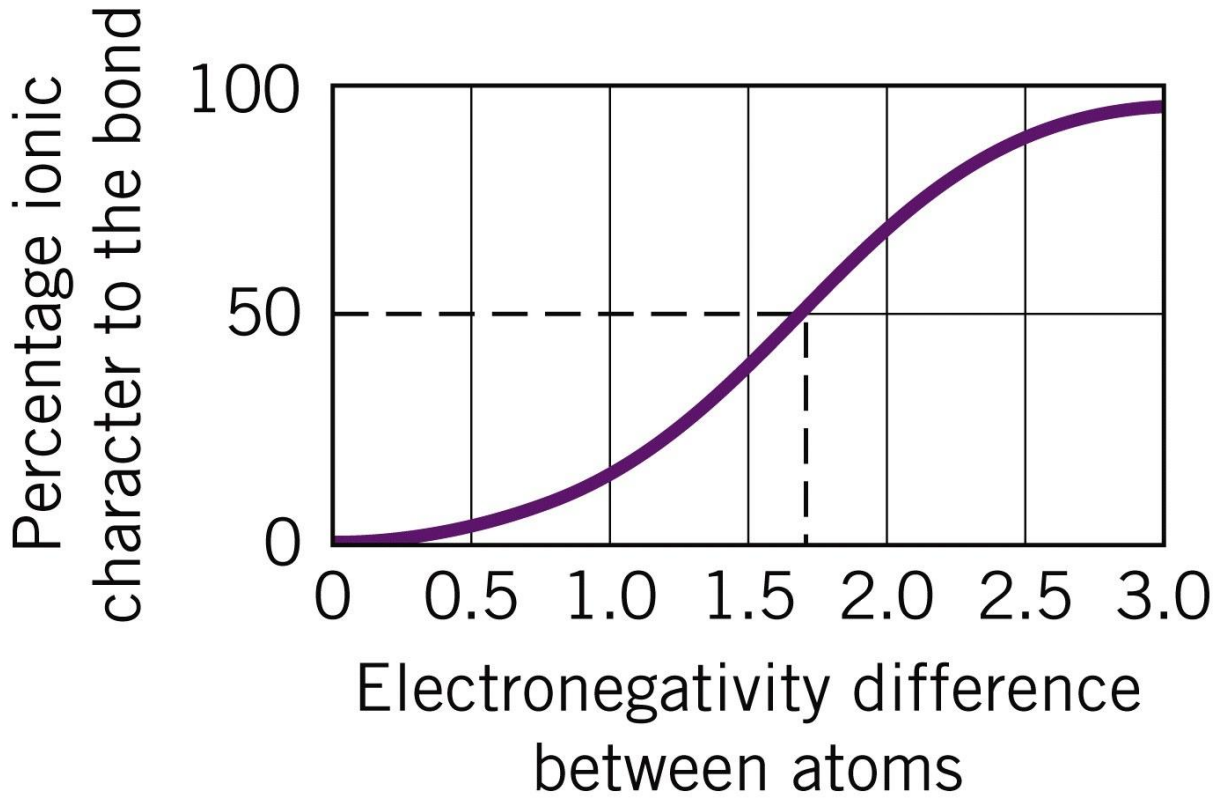
Ionic and Covalent Bonding

- 2 extremes of bonding
- Actual is usually somewhere in between.

Using Electronegativity

$$\Delta\chi = |\chi_A - \chi_B|$$

- **Difference** in electronegativity
- Measure of ionic character of bond



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Using Electronegativity

- **Nonpolar Covalent Bond**
  - No difference in electronegativity
- **Ionic Character of bond**
  - Degree to which bond is polar
  - $\Delta\chi$**> 1.7** means **mostly ionic**
    - >50% ionic
    - More electronegative element almost completely controls e^-
  - $\Delta\chi$ **< 0.5**
    - Means almost **purely covalent**
    - Nonpolar; < 5% ionic
  - **0.5 <** $\Delta\chi$ **< 1.7 polar covalent**

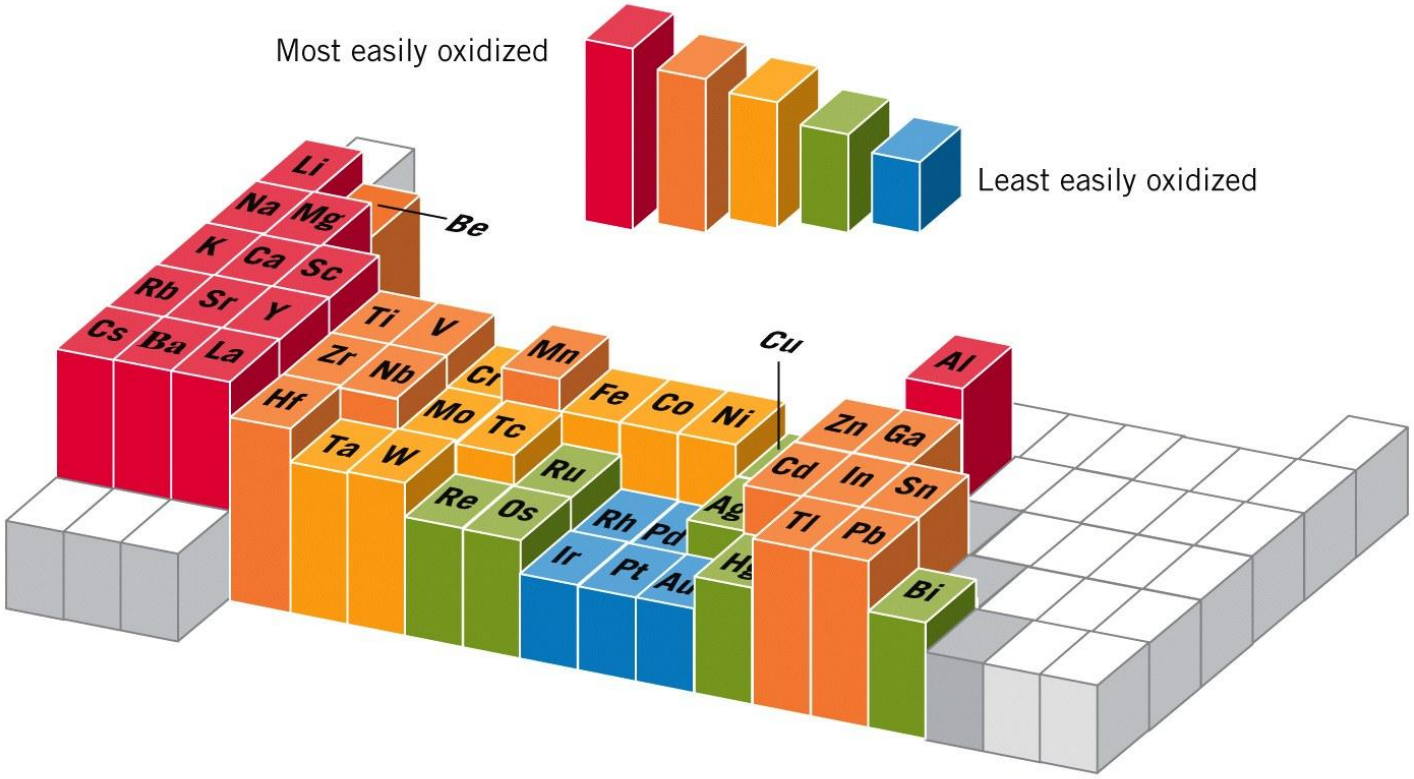
Results

- Elements in ***same*** region of Periodic Table
  - i.e., 2 nonmetals
  - Have similar χ's
  - Bonding more **covalent**

- Elements in ***different*** regions of Periodic Table
  - i.e., metal and nonmetal
  - Have different χ's
  - Bonding predominantly **ionic**

Reactivities of Elements Related to Electronegativity

- Parallels between χ and its **reactivity**
  - Tendency to undergo redox reactions



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Reactivities of Elements Related to Electronegativity

Metals

- Low χ, means easy to oxidize (groups I and IIA)
- High χ, means hard to oxidize (Pt, Ir, Rh, Au, Pd)
- Reactivity $\downarrow$ across row as $\chi \uparrow$

Nonmetals

- Oxidizing power of nonmetal (how easily reduced) parallels χ.
- Oxidizing power $\uparrow$ across row as $\chi \uparrow$
- Oxidizing power $\downarrow$ down column as $\chi \downarrow$